Egg Drop Problem: 1-3 vs. 2-2 Approach for 4 Eggs

A Mathematical Analysis

Introduction

In the classic egg drop problem, we are tasked with finding the highest floor from which an egg can be dropped without breaking, using the minimum number of drops in the worst-case scenario. Here, we analyze the scenario with $k = 4$ eggs and N floors, comparing two distinct strategies: the **1-3 approach** (the standard optimal strategy) and the **2-2 approach** (a heuristic strategy using superfloors). As $N \to \infty$, the number of drops for both approaches scales as $O(N^{1/4})$. Our goal is to determine which approach yields a better constant factor.

The 1-3 Approach (Optimal Strategy)

The "1-3 approach" corresponds to the globally optimal strategy. We choose optimal intervals to drop the first egg; once it breaks, we are left with 3 eggs, which optimally solve the remaining subproblem.

It is a well-known result that the maximum number of floors $N_k(D)$ that can be tested with k eggs and a maximum of D drops is given by the sum of binomial coefficients:

$$N_k(D) = \sum_{i=1}^{k} \binom{D}{i} \tag{1}$$

For $k = 4$ eggs, this yields:

$$N_4(D) = \binom{D}{1} + \binom{D}{2} + \binom{D}{3} + \binom{D}{4} \tag{2}$$

As D becomes very large, the term of highest degree dominates the polynomial. Thus, we can approximate the relationship as:

$$N \approx \frac{D^4}{4!} = \frac{D^4}{24} \tag{3}$$

Solving for D, we find the asymptotic number of drops required:

$$D_{1\text{-}3} \approx (24N)^{1/4} = 24^{1/4} N^{1/4} \approx 2.213\, N^{1/4} \tag{4}$$

The 2-2 Approach (Superfloor Strategy)

In the "2-2 approach," we group the N floors into N/F "superfloors," each containing F individual floors.

The strategy proceeds in two phases:

1. **Phase 1 (Finding the superfloor):** We treat the N/F superfloors as individual floors and use 2 eggs to find the critical superfloor.

2. **Phase 2 (Finding the exact floor):** Once the first egg breaks, we know the critical floor lies within a specific superfloor of size F. We then use our remaining 2 eggs to find the exact floor among these F floors.

From the general formula (1), the optimal number of drops $D_2(M)$ for 2 eggs and M floors is asymptotically:

$$M \approx \frac{D_2^2}{2} \implies D_2(M) \approx \sqrt{2M}$$

Applying this to our two phases:

- Phase 1 requires approximately $\sqrt{2(N/F)}$ drops.
- Phase 2 requires approximately $\sqrt{2F}$ drops.

The total number of drops $D(F)$ as a function of the superfloor size F is:

$$D_{2\text{-}2}(F) \approx \sqrt{\frac{2N}{F}} + \sqrt{2F} = \sqrt{2N} \cdot F^{-1/2} + \sqrt{2} \cdot F^{1/2} \tag{5}$$

To minimize $D_{2\text{-}2}(F)$, we take the derivative with respect to F and set it to zero. By the power rule:

$$\begin{aligned} \frac{d}{dF} D_{2\text{-}2}(F) &= -\frac{1}{2}\sqrt{2N} \cdot F^{-3/2} + \frac{1}{2}\sqrt{2} \cdot F^{-1/2} = 0 \\ \frac{\sqrt{2N}}{2F^{3/2}} &= \frac{\sqrt{2}}{2F^{1/2}} \\ \frac{\sqrt{N}}{F} &= 1 \implies F = \sqrt{N} \end{aligned}$$

Substituting the optimal $F = \sqrt{N}$ back into our total drops equation gives:

$$\begin{aligned} D_{2\text{-}2} &\approx \sqrt{\frac{2N}{\sqrt{N}}} + \sqrt{2\sqrt{N}} \\ &= \sqrt{2}N^{1/4} + \sqrt{2}N^{1/4} = 2\sqrt{2}N^{1/4} = \sqrt{8}N^{1/4} \approx 2.828\, N^{1/4} \end{aligned} \tag{6}$$

Conclusion

Comparing the constant factors of the highest order term $N^{1/4}$ for both approaches:

- **1-3 Approach:** $24^{1/4} \approx 2.213$
- **2-2 Approach:** $\sqrt{8} \approx 2.828$

Since $2.213 < 2.828$, the **1-3 approach is mathematically superior**. The 2-2 heuristic sacrifices efficiency for structural simplicity, leading to a constant factor that is roughly 28% larger than the true optimum.